

RAJMADAN LAKSHMANAN, M.Sc.

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Education

- 2022 – 2025  **Ph.D., Technical University of Chemnitz, Germany** in Mathematics. (*Expected*)
- 2017 – 2019  **M.Sc., University of Madras, India** in Actuarial Science.

Employment History

- 2022 – Present  **Research Associate**, Technical University of Chemnitz, Faculty of Mathematics, Chemnitz, Germany.
- 2020 – 2021  **Quantitative Analyst**, VNG Handel & Vertrieb GmbH, Leipzig, Germany.

Research Publications

Journal Articles

- 1 Rajmadan Lakshmanan and Alois Pichler. “Expectiles In Risk Averse Stochastic Programming and Dynamic Optimization”. In: *Pure and Applied Functional Analysis* 9.3 (2024), pp. 741–762.  URL: <http://yokohamapublishers.jp/online2/oppafa/vol9/p741.html>.
- 2 Rajmadan Lakshmanan and Alois Pichler. “Unbalanced Optimal Transport and Maximum Mean Discrepancies: Interconnections and Rapid Evaluation”. In: *Journal of Scientific Computing* 100.3 (2024), p. 72.  DOI: 10.1007/s10915-024-02586-2.
- 3 Rajmadan Lakshmanan and Alois Pichler. “Soft Quantization Using Entropic Regularization”. In: *Entropy* 25.10 (2023).  DOI: 10.3390/e25101435.
- 4 Rajmadan Lakshmanan, Alois Pichler, and Daniel Potts. “Nonequispaced Fast Fourier Transform Boost for the Sinkhorn Algorithm”. In: *Electron. Trans. Numer. Anal.* 58 (2023), pp. 289–315.  DOI: 10.1553/etna_vol58s289.

Preprints

- 1 Rajmadan Lakshmanan and Alois Pichler. *StochasticDominance.jl: A Julia Package for Higher Order Stochastic Dominance*. 2025. arXiv: 2502.17043 [math.OC].  URL: <https://arxiv.org/abs/2502.17043>.
- 2 Rajmadan Lakshmanan, Alois Pichler, and Miloš Kopa. *Higher-Order Stochastic Dominance Constraints in Optimization*. 2025. arXiv: 2501.14565 [math.OC].  URL: <https://arxiv.org/abs/2501.14565>.

Talks & Workshops

Talks

- 28-30 October, 2024  **Higher Order Stochastic Dominance: Advancements and Numerical Exposition**, EWGCFM & MMEI 2024, Prague, Czech Republic.
- 15-25 September, 2024  **Robust Stochastic Dominance**, EURO Summer Institute, Ischia (Naples), Italy.

Talks & Workshops (continued)

- 27-30 August, 2024  **Optimal Transport and Kernel Distances: Interconnections and Fast Computation**, 19th Doktorandentreffen der Stochastik, Brandenburgische Technische Universität Cottbus-Senftenberg.
- 20 May, 2024  **Fast Stochastic Evolution of Cells Genomics. (Poster)**, Stochastic Processes in Evolutionary Biology, Centre International de Rencontres Mathématiques (CIRM), Marseille, France.
- 30 Nov, 2023  **Entropy-regularized quantization: Theoretical Insights and Practical Applications**, Research seminar in Quantitative Finance & Statistics, Technical University of Chemnitz.
- 02 Nov, 2023  **Martingale optimal transport: Theory and applications**, PhD Seminar, Technical University of Chemnitz.
- 04-May-2023  **Stochastic Control: Monge-Kantorovich problem (Wasserstein distance) to Schrödinger bridge problem**, Research seminar in Quantitative Finance & Statistics, Technical University of Chemnitz.

Workshops

- 18-20 Sep, 2023  **Optimal transport**, Technical University of Dortmund.
- 05-16 Jun, 2023  **Optimal control of PDEs and SDEs**, Brandenburg University of Technology Cottbus-Senftenberg.

Honour & Membership

- 20-24 May, 2024  **Accommodation grant (approx. 800 €)**. Granted by Centre International de Rencontres Mathématiques (CIRM), Marseille, France.
- 2024  **Student member of Institute and Faculty of Actuaries.**
- Feb 2023 – May 2027  **Commission member of the Council of Confidence of Chemnitz University of Technology**

Skills

- Languages  English (Native or bilingual proficiency) and German (Limited working proficiency).
- Coding  Julia, Python, R, Matlab, $\LaTeX$, ...
- Misc.  Academic research, teaching, consultation and $\LaTeX$ typesetting.

References

Prof. Dr. Alois Pichler

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